

1. Answer the following questions in brief words. (4 points for each, 20 points in total)
 - (a) Two fair dice are rolled, given that fact that two dice have different values, compute the conditional probability that at least one die is 1.
 - (b) Given that $\mathbb{P}(E) = 0.9, \mathbb{P}(F) = 0.4$, prove that $\mathbb{P}(E \cap F) \geq 0.3$.
 - (c) X is a geometric random variable, prove that it has memoryless property in discrete values, that is, for any positive integers m, n , $\mathbb{P}(X = n + m | X > n) = \mathbb{P}(X = m)$.
 - (d) Rolling a fair die N times, and N follows Poisson distribution with parameter 30. Find the distribution of the number of 6.
 - (e) The joint moment generating function of X, Y is $M(t_1, t_2) = \frac{2}{2-t_1} \left(\frac{2}{2-t_2} \right)^{2022}, t_1, t_2 < 2$, find the distribution of $X + Y$.
2. (10 points) A gambler has n dollars. For each play, he wins 1 dollar with probability $\frac{1}{3}$, and loses 1 dollar with probability $\frac{2}{3}$, suppose each round of play is independent. The gambler decides to leave when he has 2023 dollars or he is broken. Find the probability that he leaves the casino with 2023 dollars.
3. (10 points) X, Y, Z are independent exponential random variables with mean 2. Compute the joint probability density function of $U = X + Y, V = X + Z, W = Y + Z$.
4. (10 points) X, Y are random variables with finite variance. If $\mathbb{E}[Y|X] = X$, show that $\text{Cov}(X, Y) = \text{Var}(X)$. Moreover, if $\mathbb{E}[Y|X] = X$ and $\mathbb{E}[X^2] = \mathbb{E}[Y^2]$, show that $X = Y$ a.s..
5. The joint probability density function of X, Y is $f(x, y) = c(y - x), x < y < x + 1, 0 < x < 2, 0 < y < 2$.
 - (a) (7 points) Find the value of c .
 - (b) (8 points) Find the marginal density function of X .

6. n men put their hats in the center of a room and mix them up, then each man randomly selects one hat. Let X be the number of men that select their own hats.

(a) (10 points) Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

(b) (5 points) Prove that

$$\frac{X - 1}{\sqrt{n}} \text{ converges to 0 in probability.}$$

7. (a) (8 points) X is a random variable with mean 0 and finite variance $\sigma^2 > 0$, prove one-sided Chebyshev's inequality: for any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\sigma^2}{\sigma^2 + a^2}.$$

(b) (4 points) Suppose the mean is 25 and the standard deviation is 4, use (a) to estimate $\mathbb{P}(X < 27)$.

(c) (8 points) X follows normal distribution with mean 25 and standard deviation 4, make use of the following table to give (i) $\mathbb{P}(X < 27)$; (ii) the value of c such that $\mathbb{P}(X \leq c) = 0.33$.

Table 5.1 Area $\Phi(x)$ Under the Standard Normal Curve to the Left of X .										
X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389